Karim Badr

London, UK | karimhbadr@gmail.com | +44 7432 251 528 | linkedin.com/in/karimhbadr | github.com/karimhbadr1

Education

University of Oxford — Oxford, United Kingdom

MSc in Psychological Research – Distinction — *Oct 2019 – Sep 2020*

- St. Hugh's College
- *Dissertation:* Uncovering Contextual and Probabilistic Contributions to Memory-Guided Behaviour

Lebanese American University — Beirut, Lebanon

BA in Psychology – High Distinction, GPA: 4.0/4.0 — *Aug 2015 – Jan 2019*

- Recipient of Full Merit Scholarship (covering full tuition fees)
- *Dissertation:* Validation of the Arabic Version of the Self-Report Altruism Scale and its Relation to Altruistic Acts in a Behavioral Paradigm

Utrecht University — Utrecht, Netherlands

Statistics Summer School – Applied Multivariate Analysis — *Aug 2018 – Sep 2018*

Work & Research Experience

SHL — United Kingdom

Senior Research Scientist — *Mar 2024 – Present*

- Science Product Owner of SHL's suite of Artificial Intelligence (AI) based assessments, owning psychometric quality across reliability, validity, and scoring for LLM- and ML-scored products
- Designing and prototyping next-generation assessment formats, including a simulation-based cognitive assessment prototype and the SVAR suite of AI-scored language assessments
- Building frameworks for AI-assisted test blueprinting and automated item generation using LLM APIs and structured prompt design; training 20+ subject matter experts (SMEs) in prompt engineering methods for item production
- Evaluating AI/ML scoring models for accuracy, robustness, and explainability: benchmarking machine scores against human raters and testing measurement invariance and subgroup DIF to support fairness and legal defensibility
- Research leader of the AI Assessments Research Program; member of the Neurodiversity & Black-Heritage Research Programs
- Leading a cross-functional team of 20+ contractors and SMEs to design new technical and language-proficiency assessments; developing R Shiny applications for routine analysis and reporting; mentoring junior researchers

Research Scientist — *Sep 2022 – Mar 2024*

- Developed and maintained SHL's suite of AI-based assessments and Next Gen solutions
- Ran validation studies (criterion-related, content, and construct) ensuring assessments are reliable, valid, and fair; analyzed millions of data points resulting in 3+ data-driven whitepapers used in global marketing and client strategy
- Consulted internal and external clients (including large banks and BPOs), delivering analytic and reporting solutions to technical and non-technical stakeholders
- Applied expertise in DEI data to provide actionable insights, including research on Neurodiversity & Ethnic minority experiences
- Ran Graded Partial Credit Models (GPCM) and other psychometric analyses; developed new content for cognitive ability, personality, and situational judgment assessments
- Collected, analyzed, and presented SHL's annual report on trends in people analytics, talent assessment, and talent management

National Foundation for Educational Research (NFER) — United Kingdom

Psychometrician / Statistician / Data Analyst — *Oct 2020 – Aug 2022*

- Produced CTT and Rasch/IRT statistics as part of the assessment development process; ran psychometric models for research projects
- Assisted in developing R packages and standard processes for automating data scoring and analysis
- Carried out multilevel model (MLM) analyses on data relating to the effect of COVID-19 on Key Stage 1 pupil performance
- Developed new standardization methods and psychometric analyses (including simulation studies) for NFER's e-Assessments
- Reviewed and updated technical manual reports for the foundation's testing products

Academic Appointments

City St George's, University of London London, United Kingdom
Honorary Lecturer, Department of Psychology *February 2026 – Present*

- Supervising MSc dissertation projects in organisational psychology, including work on an AI Readiness framework
- Collaborating with the organisational psychology group on research into AI in selection and assessment

Selected Publications

Camden, W. L., Allen, K. S., Specht, M. L., Bennett, M. W., **Badr, K. H.**, Mottram, C. L., & Gutierrez, S. L. (2024). Cognitive ability: A promising option for assessing neurodiverse talent. *Consulting Psychology Journal*, *76*(1), 20.

Tohme, P., Yaktine, N., Nassar, E., **Badr, K.**, Grey, I., & Abi-Habib, R. (2024). Exploring attachment security in a sample of Lebanese adolescents: The validation of the Arabic IPPA-R. *PLOS ONE*, *19*(3), e0298084.

Roy, P., Tang, S., Harland, J., Dirie, A., **Badr, K.**, Staunton, R., & Styles, B. (2023). Teacher Choices: The Story Time Trial. *Education Endowment Foundation, NFER, London.*

Rose, S., Twist, L., Lord, P., Rutt, S., **Badr, K.**, Hope, C., & Styles, B. (2021). Impact of school closures and subsequent support strategies on attainment and socio-emotional wellbeing in Key Stage 1: Interim paper 1. *Education Endowment Foundation, NFER, London.*

Honein-AbouHaidar, G. N., Antoun, J., **Badr, K.**, Hlais, S., & Nazaretian, H. (2020). Users' acceptance of electronic patient portals in Lebanon. *BMC Medical Informatics and Decision Making*, *20*(1), 1–12. doi:10.1186/s12911-020-1047-x

Abi-Habib, R., Wehbe, N., **Badr, K.**, & Tohme, P. (2019). Do Prisoners Mentalize Differently? Investigating Attachment and Reflective Functioning in a Sample of Incarcerated Lebanese Men. *International Journal of Forensic Mental Health*, 1–15. doi:10.1080/14999013.2019.1684403

Professional Activities

Co-lead, research community forum on AI in selection, Society for Industrial and Organizational Psychology (SIOP) Annual Conference, 2025 - 2027

Skills

Programming & prototyping: Python, R, R Shiny, SQL, Git/GitHub, MATLAB, Tableau, Unity, PsychoPy
AI/ML: LLM APIs, prompt design and engineering, automated item generation, AI/ML-based scoring, model benchmarking and explainability, algorithmic fairness auditing
Psychometrics: CTT, IRT (incl. GPCM), Rasch, factor analysis, reliability and validity evidence, DIF and measurement invariance, adaptive testing, standard setting (CEFR), equating, multilevel modelling, simulation studies
Professional standards: Standards for Educational and Psychological Testing, SIOP Principles, Uniform Guidelines, EU AI Act
Software: flexMIRT, WINSTEPS, Mplus, MLwiN, AMOS, SPSS
Languages: Arabic (Native), English (Fluent), French (Basic), Spanish (Basic)